Diffusion of Cyber Operations to State and Non-State Actors

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1 Introduction

Offensive cyber operations have become an important feature in modern conflict and in the security landscape. They have enabled the targeting of assets without military operations, enabled espionage, and blurred the line between state and non-state violence. Unlike traditional military technologies, cyber capabilities are supported by a global, decentralized digital ecosystem that enables relatively cheap, fast access to tools, knowledge, and exploits. This raises a critical question: do the properties of cyber capabilities cause them to diffuse more to state or non-state actors, and how does this vary by the type of cyber operation?

This question is important from a policy perspective, because understanding who is acquiring advanced cyber capabilities and understanding the nature of those capabilities is critical for evaluating emerging national security risks, anticipating future threats, and designing defense and resilience plans accordingly.

From an academic perspective, this question is critical to understanding the broader dynamics of technology diffusion. The cyber domain is inherently different from the physical domain, since cyber tools are often intangible, difficult to quantify, and difficult to attribute. Therefore, the extent to which theories regarding the diffusion of technology, such as lethal empowerment thoery, apply regarding cyber weapons is unclear. Examining the relationship between resource availability and power in this ambiguous domain is critical to understanding the evolving power dynamics in today's digital age.

This paper hypothesizes that offensive cyber operations that are less sophisticated, and therefore require fewer resources to execute, diffuse more extensively to non-state actors than highly sophisticated cyber operations, in accordance with lethal empowerment theory.

The analysis uses data regarding cyber incidents from the European Repository of Cyber Incidents and a measure of operation sophistication to test whether attacker type predicts the sophistication of an operation. The results reveal that non-state actors conduct a significant number of medium and high-sophistication operations, contradicting the expectations derived from lethal empowerment theory.

2 Theoretical Argument

2.1 Literature Review

The question of whether cyber operations diffuse more to state or non-state actors is guided by lethal empowerment theory and its applicability to the cyber domain. Lethal empowerment theory posits that when technologies become cheap, widely accessible, and easy to use, they tend to empower non-state actors who employ them in innovative and disruptive ways [1].

Cyber operations share many of these characteristics due to their reliance on distributed digital infrastructure rather than tightly controlled resources, and the relative ease of obtaining the tools or information required to conduct a cyber attack on online forums, such as dark web marketplaces [5]. These features of offensive cyber operations resemble earlier open technological innovations, in which broad access enabled widespread adoption by individuals and non-state groups. According to lethal empowerment theory, this diffusion should allow non-state actors to rapidly adopt and engage in offensive cyber operations.

However, cyber operations can also require significant expertise, intelligence, collection, and coordination to be effective, raising the possibility that states may still have an advantage in more sophisticated cyber operations. For example, the Stuxnet attack against Iranian nuclear centrifuges, likely conducted by the U.S. and Israel, required the development of custom exploits, technical expertise on industrial control systems and nuclear engineering, extensive coordination and integration [7]. The Stuxnet operation was likely a multi-million dollar operation, making this type of operation inaccessible to groups and actors that are not associated with nation-states [8]. Another example is the SolarWinds supply-chain compromise perpetrated by Russia, which compromised multiple U.S. government agencies. This attack involved infiltrating a software environment and using stealthy backdoors to target an organization that had multiple U.S. agencies as customers. and therefore required significant planning and technical expertise [2]. According to lethal empowerment theory, these types of highly sophisticated and resource-intensive innovations remain concentrated in the hands of nation-states and reinforce the existing power asymmetry rather than bridging it.

This broad spectrum of cyber operations, ranging from low-level operations such as website defacement to sophisticated attacks against critical infrastructure facilities, means that lethal empowerment theory cannot be applied uniformly across all types of cyber operations. On one hand, the spread of access to ransomware and exploit kits has empowered cybercrime groups, hacktivists, and other non-state actors to carry out disruptive attacks at scale. This aligns with the emphasis of lethal empowerment theory on the democratization of certain technologies. On the other hand, many attacks on critical and high-value targets require advanced techniques such as zero-day exploits and development of complex software, and are therefore carried out by nation-states or state-affiliated actors.

Additionally, a potential weakness of lethal empowerment theory when applied to the cyber domain may arise from the fact that it is strongly rooted in explaining the diffusion of physical technologies such as dynamite and the Kalashnikov assault rifle [1]. There is some application of this theory to Internet-connected technologies and information sharing, but cyber capabilities may not always depend on freely available code and knowledge, and may depend on tightly controlled intelligence and access to closed networks.

The diffusion dynamics of cyber capabilities are also inherently different from that of physical capabilities. Offensive cyber capabilities may or may not be resource-intensive to create, but they are often easy to replicate. Additionally, the cyber domain is also characterized by a "use it or lose it" dynamic, where vulnerabilities to be exploited, such as zero-day vulnerabilities, often lose their value once they have been discovered and patched [10]. This complicates the issue of "accessibility" in the cyber domain, since highly sophisticated tools diffuse rapidly once developed, but sophisticated ransomware and malware may be limited in its diffusion depending on the vulnerabilities it is based on.

These properties of cyber weapons complicate the expected outcomes according to lethal empowerment theory. Although the theory predicts that accessible tools empower non-state actors, it can be difficult to define accessibility and broadly classify whether cyber weapons are "accessible" or not.

2.2 Hypothesis

The main question guiding this analysis is whether the sophistication of a cyber attack varies based on the type of actor responsible for executing the attack.

Based on lethal empowerment theory, the hypothesis is that non-state actors will be more likely than state actors to conduct less sophisticated offensive cyber operations. The logic behind this expectation is that non-state actors will generally lack the financial resources, intelligence, and advanced technical and domain-specific expertise to develop and deploy advanced offensive cyber ca-

pabilities, especially those that are capable of large-scale disruption of critical targets. Instead, they will tend to rely on publicly available tools, leaked exploits, and relatively inexpensive malware and ransomware. This means that non-state actors should mainly be conducting less sophisticated operations such as website defacement, phishing operations, basic ransomware, credential theft, etc, since they require less technical expertise and the knowledge and resources to execute these attacks is often circulating on dark web marketplaces and and online forums. This is aligned with lethal empowerment theory, since non-state actors should mainly be rapidly adopting cheap and accessible technology.

In contrast, sophisticated cyber operations are expected to mainly be executed by state or state-affiliated actors. This is because complex, multi-step operations that require long-term intelligence collection, undisclosed vulnerabilities, and extensive coordination are resource-intensive and should therefore be relatively inaccessible to non-state actors. For example, attacks that target critical infrastructure such as a power grid require domain specific technical expertise regarding industrial control systems, substations, and power systems, as well as knowledge about security vulnerabilities and exploit development.

3 Methodology

To evaluate whether offensive cyber operations diffuse more to non-state actors than to state actors, and whether the sophistication of a cyber operation is related to their diffusion, I conducted a statistical analysis using incident-level data from the European Repository of Cyber Incidents (EuRepoC) [3]. The EuRepoC dataset is a comprehensive data set of cyber incidents based on verified public reporting. For each attack, it includes information about the attacker, the target, the type of incident, technical characteristics, and impact, when the information is known. This dataset is well-suited for this analysis because it provides information on both the actors involved in the operation, and broadly classifies the actors as well, which is useful for evaluating attacks based on whether they are associated with state or non-state actors. Additionally, this data set provides a foundation for evaluating the sophistication of cyber attacks since it also records a "cyber intensity" score based on the nature of the cyber attack, and also records whether a zero-day vulnerability was used. The sophistication of cyber attacks is a metric that is necessary for this analysis, but it is relatively difficult to quantify, so the EuRepoC data provides sufficiently detailed information to approximate the sophistication of an attack.

The objective is to determine whether non-state actors are more or less likely than state actors to conduct cyber operations that are highly sophisticated.

3.1 Data Preprocessing

First, the attackers responsible for each cyber attack were classified as being either a "state" or "non-state" actor in a binary attacker variable based on the categories provided in the EuRepoC dataset. The EuRepoC data distinguishes between multiple types of attackers, such as 'State', 'Non-state groups', 'Individual hackers', 'Other', and 'Non-state actor, state-affiliation suggested'. For binary coding, attackers in the 'State' and 'Non-state actor, state-affiliation suggested' were coded as state actors and given a value of 0. Actors in the 'Non-state group', 'Individual hackers', or 'Other' categories were coded as non-state actors and given a value of 1. The cases in which the attacker was unknown were excluded from the analysis. This coding reflects the fact that groups that may officially be considered "non-state" groups have resources and capabilities close to that of a formal state agency if they are backed and funded by nation states. Therefore, only cybercriminal organizations, hacktivist groups, and groups without confirmed state support were included as 'non-state actors' in the analysis.

Second, a sophistication measure for cyber operations was created to classify various cyber operations based on their complexity. Using the theoretical framework of lethal empowerment theory, the sophistication metric was approximately a function of the technical difficulty and the resources required. The sophistication metric included the following components:

1. Weighted Cyber Intensity: This variable was directly included in the EuRepoC data set. It captures the severity of the cyber incident by combining the technical effects (such as disruption, data theft, or physical consequences) with the strategic importance of the target. A score is assigned to the technical effects of the attack, and then this unweighted score is multiplied by a target multiplier that assigns a higher weight to attacks that lead to the disruption of critical services. This can include targets such as critical infrastructure and military entities.

2. Use of Zero-Day Vulnerability: The second component of the sophistication score was whether the attacker used a zero-day vulnerability. The use of zero-day exploits in a cyber attack are a strong indicator of an attacker's capabilities and overall attack sophistication because they require access to undisclosed vulnerabilities that are hard to access and require significant expertise or financial resources to obtain. In the EuRepoC data, the use of a zero day vulnerability is simply coded as "Yes" or "No", indicating whether the attack involved a confirmed zero-day. To reflect the significant increase in technical difficulty associated with obtaining and using a zero-day vulnerability in a cyber attack, this component was given a weight of 3 when a zero day was used and 0 otherwise. This ensures that operations involving zero-day vulnerabilities meaningfully increases

the operation's sophistication classification, and ensures that a zero-day operation is never classified as having low sophistication.

The sophistication score was created by adding the weighted cyber intensity score and the zero-day score.

$$\text{Sophistication Score} = \text{Weighted Cyber Intensity} + \text{Zero-Day Score}$$

This metric captures a range of operation types from low-impact attacks with no advanced techniques to high-impact operations involving significant impact and the use of unknown vulnerabilities.

Finally, the continuous sophistication score was converted into an ordered categorical variable with three levels: low, medium, and high.

1. Low Sophistication: Operations that had a combined sophistication score of two or below were classified as having low sophistication. These attacks had relatively low impact and complexity, and do not involve the use of zero-day vulnerabilities. An attack under this classification is the November 2025 breach of South Korea's top crypto exchange, Upbit, where attackers exploited a vulnerability in a cryptocurrency hot wallet and stole digital assets. Although there was financial impact, the operation was not extremely technically complex, such as involving advance coordination or capabilities.

2. Medium Sophistication: Operations that had a combined sophistication score between three and six were classified as having medium sophistication. These operations involved moderate disruption or the use of a zero-day vulnerability. They generally require a higher degree of technical skill and planning, such as disruptive ransomware attacks targeting important facilities or multi-step data exfiltration attacks. An example of an attack under this category was the ransomware attack on Wayne Memorial Hospital in 2024, which disrupted operations, encrypted medical data, and exposed sensitive personal information for more than 160,000 people. The attack involved multiple steps and had a significant impact. This attack had a sophistication score of six, and is therefore close to the upper threshold for attacks classified as having a medium sophistication. Another attack under this category was the GRU-linked breach of the Swedish Sports Confederation by Russia-linked hackers between 2017 and 2018. This attack involved targeted data exfiltration to expose information of Swedish athletes. The attack had a sophistication score of three, and is therefore close to the lower threshold for attacks classified as having a medium sophistication. The attack involved the use of compromised accounts and exfiltration, so it involved some targeted access and coordination, but the overall complexity and impact was not high.

3. High Sophistication: Operations that had a combined sophistication score of seven or higher were classified as having high sophistication. These attacks either had a high level of impact, such as the targeting of critical infrastructure, or combined a moderate impact attack with the use of a zero-day vulnerability. An example of an attack under this category was the 2025 attack on the Crisis24/OnSolve emergency alert system, which disrupted emergency notification services across multiple U.S. municipalities. This incident had a high sophistication score because it caused significant and prolonged disruption to critical infrastructure.

3.2 Statistical Analysis

To evaluate whether non-state actors are more likely than state actors to conduct lower sophistication cyber operations, I used a logistic regression model using the sophistication variable as the dependent variable. Since the sophistication measure is an ordinal variable, with three levels, i.e., low, medium, and high, a logistic regression model can estimate how attacker type influences the likelihood of an incident being classified into each sophistication category. The attacker type is coded as '0' for state actors and '1' for non-state actors. The model is as follows:

$$\text{sophistication} = \beta_0 + \beta_1(\text{attacker type}) + \epsilon$$

The model estimates how the log-odds of a cyber attack being in a higher sophistication category changes when the attacker is a non-state actor rather than a state actor. In the model, the effect of the attacker type on the likelihood of an incident being more sophisticated is the same for a shift from low to medium sophistication and from medium to high sophistication.

4 Results

Based on the data, 1854 cyber attacks were classified as low-sophistication, 2014 as medium-sophistication, and 44 as high-sophistication. Therefore, very few attacks involve a meaningful disruption of critical facilities or complex, multi-step attack methods. The model's threshold coefficients also support this pattern. The first threshold (estimate = 0.716, odds ratio = 2.05) shows that cyber attacks are approximately twice as likely to fall above the low-sophistication category, which is consistent with the dominance of medium-level attacks in the data. The very large second threshold (estimate = 7.483, odds ratio = 1,778.76) indicates how difficult it is for an incident to be categorized as highly sophisticated. Additionally, the data shows that the number of non-state cyber operations is more than that of state cyber operations. This difference is most significant after 2022, when the number of non-state operations sharply increases.

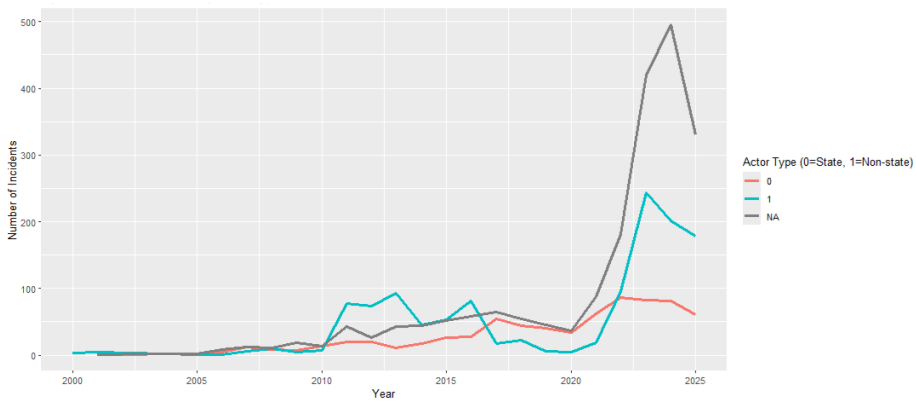

Figure 1: Cyber Incidents over Time Over Time by Actor Type

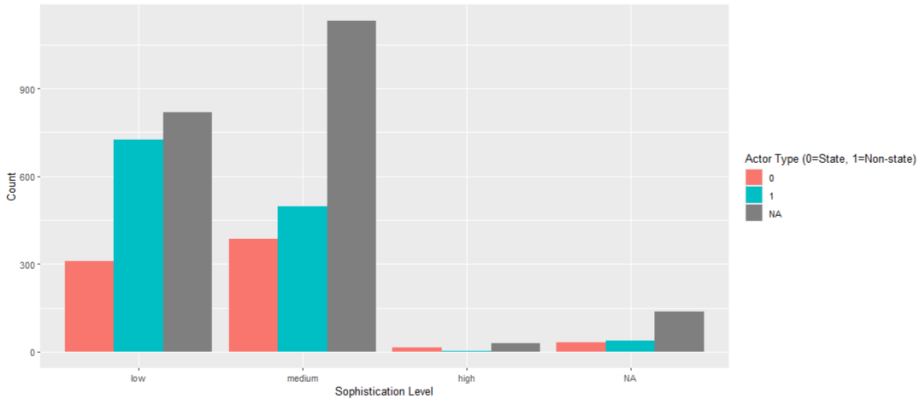

Figure 2: Cyber Attack Sophistication by Attacker Type

	Estimate	Odds Ratio	Std. Error	t value	p value
attacker_type	1.6231932	5.069252	0.09981894	16.26137	1.855284e-59
low | medium	0.7162746	2.046794	0.07809736	9.17156	4.662234e-20
medium | high	7.4836727	1778.761637	0.50756209	14.74435	3.345285e-49

Table 1: Ordinal Logistic Regression Results

The output from the logistic regression model provides clear evidence of how the type of attacker is associated with the sophistication of offensive cyber operations. The model estimates an odds ratio of 5.06 for *attacker type*, indicating that non-state actors are approximately 5 times more likely than state actors to conduct a higher sophistication attack, when other factors are held constant. The small p-value of 1.8×10^{-59} indicates that this relationship is statistically significant, and therefore an attacker being a state or non-state actor is a strong

predictor of how sophisticated a cyber attack is.

These results indicate that the type of attacker, i.e., state or non-state, is significant in shaping the complexity of a cyber operation. Additionally, non-state actors disproportionately conduct attacks of higher sophistication.

5 Discussion

The statistical results directly contradict the earlier hypothesis, i.e., that non-state actors conduct less sophisticated attacks than state actors due to a difference in resources. Lethal empowerment theory anticipates that when disruptive technologies become cheap, widely accessible, and easy to use, non-state actors will increasingly adopt them, and in the cyber domain, this seems to correspond to cyber operations at lower levels of sophistication. Highly sophisticated and impactful attacks were hypothesized to require high-end capabilities, which require substantial expertise, intelligence resources, and organizational capacity, and were expected to remain concentrated within states.

However, the empirical results indicate that non-state actors are actually more likely than state actors to conduct medium and high sophistication attacks. The odds ratio of 5.06 indicates that non-state actors are disproportionately represented in offensive cyber operations involving the use of advanced vulnerabilities and causing significant disruption. Rather than being concentrated at the lower end of sophistication levels, non-state actors seem to exhibit capabilities that are predicted to be concentrated in the hands of states, according to lethal empowerment theory.

One potential explanation is that the cyber domain is inherently more democratic compared to previous "closed" technological revolutions regardless of attack sophistication, and so lethal empowerment theory still holds.

Under this explanation, the characteristics of the cyber domain and offensive cyber tools and techniques are different from previous technologies in the physical domain. For example, cyber operations do not necessarily rely on expensive or specialized physical infrastructures or state-controlled resources. They rely significantly on knowledge-based resources rather than hardware-based resources, such as coding expertise and the development of vulnerabilities. Additionally, the cybercrime domain is structured such that advanced tools and exploits may be expensive, but are still relatively open due to commercial availability of advanced tools and exploits over the dark web to any group with sufficient funds.

One way these characteristics of cyber operations have appeared is in the fact that sophisticated state capabilities have diffused to non-state groups in the past, which reduces the resource and capability gap between state and non-state

actors. For example, the tool EternalBlue was originally created by the U.S. National Security Agency, but was picked up by non-state actors once they were leaked, and used in the WannaCry cyber incident [9]. Therefore, the dynamics of technological diffusion is fundamentally different in the cyber domain, where knowledge itself is a resource, and can be replicated and distributed at almost no cost once it exists. When advanced exploits are released either intentionally or by accident, even highly complex capabilities become accessible to actors with few resources.

Another theoretical explanation for the disproportionate representation of non-state groups in attacks of higher sophistication is the differing goals between states and non-state groups. States may often avoid highly disruptive cyber operations, especially those that disrupt critical infrastructure and significantly affect civilians, due to concerns about escalation and retaliation. Therefore, the most sophisticated cyber operations which disrupt critical targets may be seen more among non-state actors than state actors due to restraint rather than a lack of capability. Non-state actors, on the other hand, are not generally constrained by these strategic concerns. However, this only explains why state actors do not seem to be conducting more highly sophisticated attacks and not why non-state actors are able to conduct relatively sophisticated attacks despite resource constraints. Therefore, strategic considerations are not sufficient to explain the empirical results.

Therefore, these findings indicate that lethal empowerment theory is mostly able to explain the diffusion of offensive cyber capabilities, and that the initial hypothesis was incorrect since it did not correctly characterize the cyber domain. The theory correctly predicts the widespread adoption of cyber capabilities by non-state actors, even for more sophisticated operations. This means that high-end cyber tools, even if expensive to develop, diffuse widely to non-state actors due to the relatively low cost of replication and distribution. In the cyber domain, this means that non-state actors become empowered to execute both low-level and complex, impactful attacks that may require advanced capabilities and technical expertise. In this sense, the results reinforce the core idea behind lethal empowerment theory, i.e., that when barriers to the acquisition and use of a tool are reduced, they are rapidly adopted by actors with fewer resources, and therefore reshape the balance of power.

6 Limitations

A critical characteristic of cyber capabilities is that an actor's capability is generally unknown or "invisible" until it is used. Traditional military capabilities, especially military technology, are generally observable, since they often rely on physical equipment, training exercises, etc. However, the cyber domain is inherently covert. For example, the existence of a zero-day vulnerability or

sophisticated malware is unknown until it is used and identified. Therefore, states may have highly advanced cyber capabilities that simply do not appear in datasets like the EuRepoC dataset because they have not been deployed or are used only in covert operations where attribution is difficult or impossible.

Additionally, the distinction between cyber attacks that are categorized as having "medium" and "high" sophistication is somewhat dependent on the amount of disruption and impact that the cyber attack had in this analysis. However, many cyber operations conducted by states may aim to be stealthy, due to the goal being persistence and intelligence collection rather than large-scale disruption. These operations are still sophisticated and difficult to execute. However, they would not score as highly on a sophistication metric that accounts for impact, like the metric used in this analysis.

These characteristics of the cyber domain highlight a key limitation of this analysis and provide an alternative explanation for the finding that non-state actors are disproportionately represented in sophisticated cyber attacks. Since non-state groups have different goals in conducting cyber operations than state actors, such as financial goals or goals related to political signaling, they often conduct operations that are much more visible and attributable than state actors. States may deliberately avoid high-impact or highly-visible attacks for strategic reasons. Therefore, the relatively low number of state-affiliated cyber attacks compared to non-state attacks may be due to a lack of reporting, and therefore skew data and results towards non-state actors. This is also supported by the fact that a significant number of cyber incidents in the data have not been attributed to an attacker at all.

Additionally, this highlights an important limitation of analyses based on incidents in the cyber domain. Incident-level data mainly capture operations that are detected and attributed, which biases the data towards visible and disruptive operations. Therefore, statistical results may reflect a difference in the visibility of operations rather than a difference in capabilities.

Finally, an important consideration is that the sophistication metric used in this analysis, which used weighted cyber intensity and the use of zero-day vulnerabilities, does not fully measure attack sophistication. For example, some attacks by non-state actors with significant impact may be due to the exploitation of known vulnerabilities that haven't been patched or due to poor cybersecurity posture and defenses. This would reflect timing and opportunity rather than genuinely advanced capabilities. For example, the NotPetya ransomware attack had a significant impact on the shipping company Maersk, affecting global supply chains. However, Maersk was found to have multiple unpatched vulnerabilities and outdated operating systems, contributing to the large-scale impact [4]. However, this attack would be represented on the upper end of the sophistication metric due to its impact. Therefore, a more robust sophistication metric that includes a detailed analysis of current techniques and known attack path-

ways, perhaps using the MITRE ATT&CK framework, is necessary to more conclusively evaluate the relationship between cyber attack sophistication and adoption [6].

7 Conclusion

The central question motivating this analysis was whether offensive cyber operations diffuse more extensively to non-state or state actors, and whether this varied based on the sophistication of the attack. Applying lethal empowerment theory to offensive cyber operations, the initial hypothesis predicted that non-state actors would conduct primarily low-sophistication cyber operations, since these require fewer resources, less expertise, and rely on widely accessible tools. In contrast, state, which were assumed to have more technical, financial, and intelligence capabilities, were expected to be significantly associated with highly sophisticated cyber operations.

However, the empirical findings contradict this expectation. The analysis shows that non-state actors are more likely than state actors to carry out medium and high-sophistication operations. These initial results suggest that in the cyber domain, the barriers associated with developing advanced offensive capabilities, such as advanced technical expertise and intelligence are not as significant as previously assumed.

These results have several implications. First, they indicate that lethal empowerment theory is broadly applicable to the cyber domain. Accessible cyber tools diffuse widely and to non-state actors, which is posited by lethal empowerment theory, and even sophisticated cyber capabilities seem to diffuse due to certain structural characteristics of cyberspace. These characteristics include the ease of replicating cyber capabilities once they are created and the relative openness of attack knowledge and exploits.

For policymakers, these results highlight that non-state cyber actors are an increasingly important consideration in the security landscape. Therefore, it is increasingly important to adopt stronger cybersecurity measures in order to defend against the proliferation of offensive cyber capabilities

However, the analysis has some key limitations, and so the results are not entirely generalizable without more robust data and metrics. The incident-level data only account for known cyber operations and not covert or unattributed ones. Additionally, the sophistication metric may overestimate the impact of a cyber attack and not truly quantify the resources required to develop and execute a cyber attack.

Therefore, this analysis provides an initial indication that the diffusion of offen-

sive cyber capabilities, across all levels of sophistication, is broadly consistent with the theory of lethal empowerment. However, generalization is limited by the fact that the cyber domain can be highly covert, and that a more robust metric to measure sophistication and required resources for attack execution is necessary.

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